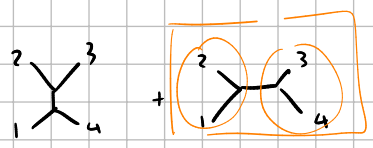
$$f(z) = \boxed{\frac{a_{-1}}{z}} + a_0 + a_1 z + a_2 z^2 + \cdots$$

$$\mathrm{Res}\, f(z) = a_{-1}$$

$A_4 =$ 

$S^\dagger = S$

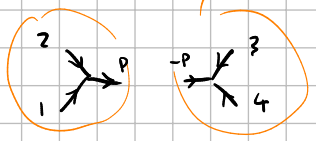
$\Rightarrow$ residue factorization

$$= \frac{1}{s_{14}} + \frac{1}{s_{12}}$$

$O(1)$ w.r.t. s_{12}

4-momentum $p^0 = E(\vec{p})$

$\mathrm{Res}_{s_{12}=0} A_4 \propto 1$

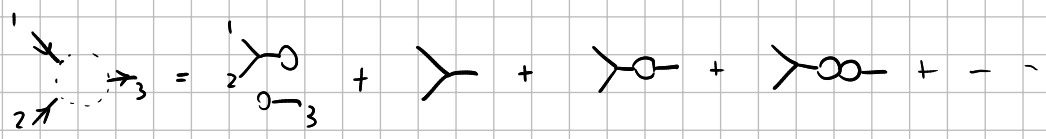


A_3 A_3

---

non-local: the QFT cannot be built with path integrals of an action

(canonical quantization starting from Lagrangian)

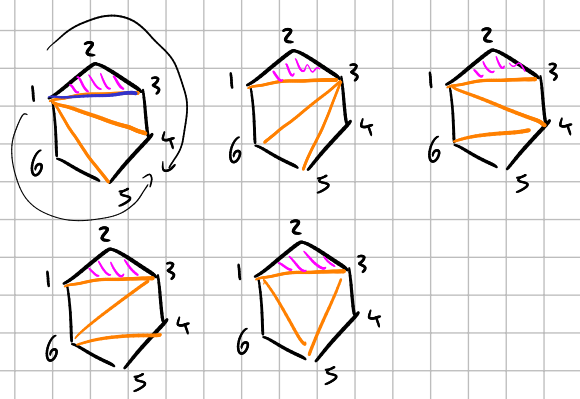


$$f(z,w) = \frac{1}{z} + \frac{1}{w} + \frac{z}{w^2}$$

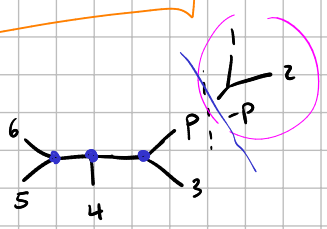
expand in powers of $\frac{1}{z}$ $\qquad f(z,w) = \frac{1}{z} + O(1) \Leftrightarrow \mathrm{Res}\, f_{z=0} = 1$

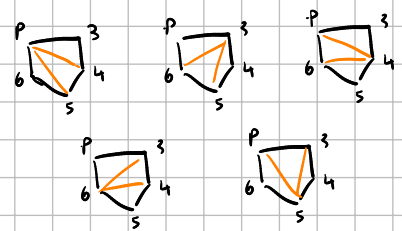
$\mathrm{Res}(A_6)$ look at residue at x_{13}

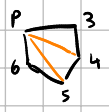
$$A_6 = \frac{1}{x_{13}}\left(\boxed{\frac{1}{x_{14}x_{15}} + \frac{1}{x_{36}x_{35}} + \frac{1}{x_{14}x_{46}} + \frac{1}{x_{36}x_{46}} + \frac{1}{x_{35}x_{15}}}\right) + O(1)$$

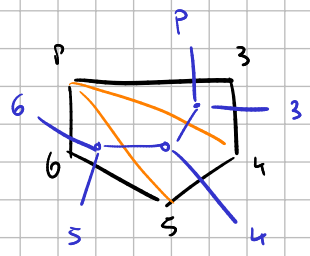


$$\frac{1}{(P_4+P_5+P_6)^2\,(P_5+P_6)^2}$$





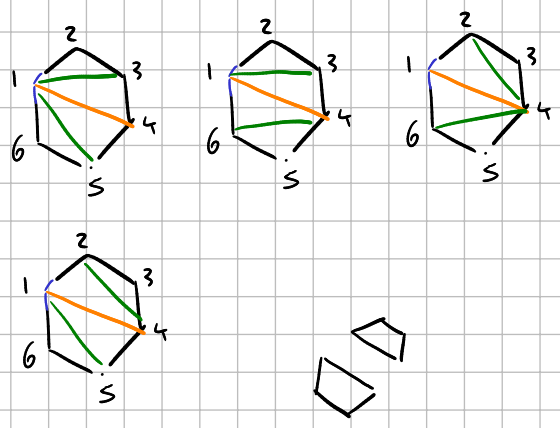




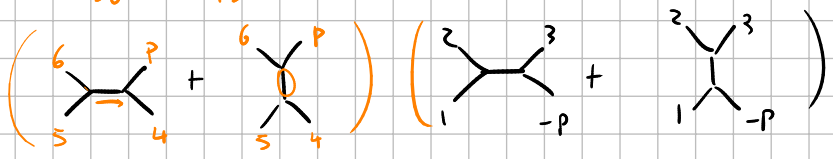
$$A_6 = \frac{1}{x_{14}}\left(\frac{1}{x_{13}\,x_{15}} + \frac{1}{x_{13}\,x_{40}} + \frac{1}{x_{24}\,x_{40}} + \frac{1}{x_{24}\,x_{15}}\right)$$

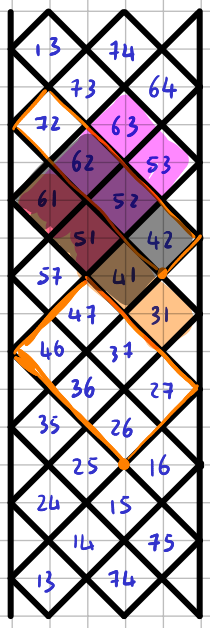
$$= \frac{1}{x_{14}}\left(\frac{1}{x_{15}}\left(\frac{1}{x_{13}} + \frac{1}{x_{24}}\right) + \frac{1}{x_{40}}\left(\frac{1}{x_{13}} + \frac{1}{x_{24}}\right)\right)$$

$$= \frac{1}{x_{14}}\left(\left(\frac{1}{x_{15}} + \frac{1}{x_{46}}\right)\left(\frac{1}{x_{13}} + \frac{1}{x_{24}}\right)\right)$$



$s_{56} + s_{45}$





→ 1 zero

$c_{42} = c_{52} = c_{62} = c_{72} = 0$

→ 2-zero

$c_{26} = c_{36} = c_{46} = 0$

$c_{27} = c_{37} = c_{47} = 0$

$\rightarrow$ $c_{6j} = 0$, $c_{7j} = 0$ $\quad j = 9, \dots, 13$

$c_{14} = c_{15} = c_{16} = 0$

$c_{24} = c_{23} = c_{26} = 0$

$1\,2\,3\,4\,5\,6\,7 \rightarrow 6\,7\,1\,2\,3\,4\,5$

$c_{62} = c_{63} = c_{64} = 0$

$c_{72} = c_{73} = c_{74} = 0$

<u>1 zero:</u> $\quad c_{1j} = 0 \quad j = 3, \dots, n-1$

2-zero $\quad \begin{cases} c_{1j} = 0 \\ c_{2j} = 0 \end{cases} \quad j = 4, \dots, n-1$

3-zero $\quad \begin{cases} c_{1j} = 0 \\ c_{2j} = 0 \\ c_{3j} = 0 \end{cases} \quad j = 5, \dots, n-1$

$$\boxed{\underline{k\text{-zero:}} \quad \begin{cases} c_{1j} = 0 \\ c_{2j} = 0 \\ \vdots \\ c_{kj} = 0 \end{cases} \quad j = k+2, \dots, n-1}$$

$\sum_j c_{1j} = 0$ with 1-zero $\Rightarrow$ $S_{12} = -S_{1n}$

$$c_{ij} = X_{ij} + X_{i+1\,j+1} - X_{i+1\,j} - X_{i\,j+1}$$

$n = 7$ $\quad k = 3$: $\quad j = 5, 6$

$c_{15} = c_{16} = 0$

$c_{25} = c_{26} = 0$

$\vdots$

$c_{75} = c_{76} = 0$

$1 \le i < j \le n$

$$X_{ij} = \left(\sum_{k=i}^{j-1} p_k\right)^2$$

Q: which X_{ij} are affected by setting $c_{1j} = 0$ for $j = 3, \dots, n-1$?

$$c_{ij} = -(p_i + p_j)^2$$

$$c_{1j} = -(p_1 + p_j)^2 = -2 p_1 \cdot p_j$$

$X_{1\,j+1}$ depends on p_1

$$\| \quad \left(\sum_{k=1}^{j} p_k\right)^2 = (\overbrace{p_1 + p_2}^{A} + \overbrace{\cdots + p_j}^{B})^2 \qquad A^2 + 2AB + B^2$$

$$= \underbrace{(p_1 + p_2)^2}_{S_{12}} + \underbrace{2(p_1 + p_2)\cdot(p_3 + \cdots + p_j)} + \underbrace{(p_3 + \cdots + p_j)^2}_{S_{3,4,\dots j}}$$

$$2(p_1 + p_2)\cdot(p_3 + \cdots + p_j) = 2p_1 \cdot (p_3 + \cdots + p_j) + 2p_2(p_3 + \cdots p_j)$$

$$= \sum_{i=3}^{j} \underbrace{2p_1 \cdot p_i}_{S_{1i}} + \sum_{i=3}^{j} \underbrace{2p_2 \cdot p_i}_{S_{2i}}$$

$$S_{i_1 i_2 \cdots i_k} = (p_{i_1} + p_{i_2} + \cdots + p_{i_k})^2$$